

A YEAR WITH THE COOPER M21 .204 RUGER



By John A. Del Savio

August 6, 2007 The Rifle Range, Pine Tree Rifle Club, Johnstown, New York

Over 20 shooters were at the benchrests making ready for the evenings rimfire benchrest match when Gary Steding arrived. He and I have been friends for many years hunting together and shooting in competition, we know each other's attraction to certain firearms. This familiarity has evolved into a game we play when certain firearms are located for sale. It's a simple game of informing the other of said firearm found, and watching the other guy try to fight off the urge to buy it. Most of the time we can't, and end up buying the firearm. At this point in the game I have the lead over Gary of about 4 to 1.

Gary had just returned from Albany, NY where he stopped at a gun shop that is a Cooper Firearms dealer. Gary knows that I have a weak spot for Cooper Rifles, and with a nonchalant manor he told me of a Cooper M21 in Ruger .204 that was for sale. I could tell by the look on his face that he was enjoying every minute of what he was doing and using the skill of a high dollar defense attorney playing to the jury. Most of the shooters on the firing line picked up on what was happening and joined in with Gary playing on my weakness for Coopers. Jr Dean piped up with "What's the wood look like?", and Gary replied with "Outstanding piece of tiger stripped walnut that is figured from the butt to the tip of the forearm, and it's also left handed".

Three benchrest to my left sat Fran Plew who also plays the game that Gary and I play. I'm in the lead with him also about 7 to 1 in my favor. Fran pipes up with "I'll ride down to Albany with you in the morning when you pick it up". Even Dr. Richard Maretzo, my friend and shooting mentor, whom I hold in the highest of regard took part in setting me up! The conspiracy was more than obvious!

If I had been tied to a foremast as Ulysses was as his ship passed the Island of Lesbos I might have had a chance, but there were no restraints to help me control myself. They had me and my fate of purchase was sealed!

August 7, 2008 10:00 AM

Fran Plew and I made our way to Albany, and talked about his experience with his Cooper 21 in .204 Ruger. Fran bought one of the first Cooper 21s in 204 that was available locally back in 2005. Since he has used it to take several woodchucks and tons of prairie dogs in South Dakota in 2006 and again in 2007.

In South Dakota Fran did some impressive shooting with it making several kills in excess of 400 yards. Anything that came into his scope 300 yards and under was in serious trouble! I believe that the longest shot he made was just over 500 yards. Even more impressive was the fact that Fran was shooting factory ammo for most of the 2006 P-dog trip! In 2007 Fran used the .204 as his main rifle on our South Dakota prairie dog shoot. On that trip Fran gave me the chance to shoot a few p-dogs with that rifle. Right from the start it was a pleasure. There was very little recoil, a very flat trajectory, and I was able to maintain the sight picture under recoil. I was nearly sold right there. I guess that is where the seed was planted for me to own a Cooper 21 rifle in .204 Ruger.

10:30 AM Fran and I entered B&J Shooter Supply in Albany, New York. There on the rifle rack where displayed about a dozen Cooper rifles. I scanned the rack's inventory for the rifle Gary spoke of, but could not locate it. Then store owner Brian Olesen appeared holding the left handed Cooper 21 in .204 Ruger. He had a big smile on his face and said "I'm sure you're looking for this". Gary had really done a great job of setting me up for this one!

This Cooper was everything that Gary had said it was. The wood was outstanding and stock fit like it was made for me. A price was agreed upon, money exchanged hands, and I was on my way home with a warm fuzzy feeling.

Once home a Leupold 6.5-20X scope was mounted and orders were made to acquire loading dies, bullets, brass, and all the special little tools needed for working with .204 diameter cartridges. The first test firing of the new Cooper would have to wait until all necessary components were gathered, and test cartridges assembled.

One would think that being a hand loader for 44 years I would have nearly all needed to start loading for the new rifle. No, I didn't! For those of you that will venture into the realm of the .204 caliber I have made a list of the tools that you will need before you start loading.

Here are a few items you will need, .204 case trimmer pilot, .204 neck turning pilot, a pilot for the inside flash hole chamfering tool, and a powder funnel that will work with that wee little cartridge case neck. Standard powder funnels are just too large and will not work with the .204.

Loading tiny little bullets straight can be difficult, so I ordered a set Redding Competition dies. They have a precision straight line bullet seating die that makes loading those tiny bullets easy and precise. I also like the Wilson straight line seating die, and you will not go wrong with it either. The Wilson die is a little slower, but their precision is unquestionable.

Hold it right there before you call that order in. You are also going to need items to clean that rifle with. Your cleaning rod that works in your 22 caliber bore is not going to work in that .204 barrel. It's simply too big. You are also going to need bore brushes, cleaning jag, bore guide, and small cleaning patches.

A .204 cleaning brush is not as robustly designed as a .22 caliber bore cleaning brush. They wear out very quickly, and ordering them a dozen at a time is not over stocking.

A word of caution about cleaning patches is necessary. This is a small bore and getting a cleaning patch through it that is made for a .22 caliber can be very difficult. They are in my opinion just too large. They can get stuck, and you can put a bend in the cleaning rod trying to push it through forcibly. I highly suggest that you make purchase of 1" round cleaning patches. This will save you a pile of grief!

Mid - August had passed when the loading tools and components arrived. As all you handloaders know it takes time to prep 200 new cases and develop a load. I was running out of time to take the new rifle to the field on a chuck hunt.

In upstate New York the woodchucks (ground hogs) start spending more time underground than above by the end of the second week of September, and I wanted to get in a few trips before they went into hibernation. Fran Plew had developed a load for his .204 Ruger using Hornady 40 grain V-Max bullets, BL-C#2 powder and Federal 205M primers. This load was shooting accurately and near 4,000 fps in his Cooper 21. So cartridges were assembled with Fran's load data and the rifle sighted in for a 100 yard base zero. Accuracy with this load was acceptable to get started with and the rifle was pronounced field ready.

The first trip to the field with the new Cooper was late in August on a day that was a pleasure to be a field. Blue skies, temperatures in the 70s F, and a light breeze made the day very comfortable. My hunting partner for the day was Dr. Richard (Doc) Maretzo who was as anxious to see the .204 Ruger perform as I was. The area we hunted is typical of upstate New York dairy farm country with rolling hills and fields that can be small to hundreds of yards long. With all the rain we had throughout the summer farmers were late cutting their hay fields, and we were limited to only a few fields that were in condition suitable to hunt.

Scouting from ATVs we found a back field that was cut clear of hay, and shaped nearly as a Roman amphitheater. At the bottom of the half bowl field were a few dozen round hay bales that were wrapped in white plastic covers. Behind them was a brush covered creek bed that led to the base of a steep wooded hill of several hundred feet. The hay bales made a perfect blind to hide the ATV and provide hunters hide. Maximum range in the field was just a little over 300 yards, and any varmint in the field would be well within range of the rifles we carried.

It was the perfect spot. The sun was at our back, and we had a vantage point to put anything in our sights that may appear while remaining concealed. Not long had passed when chuck #1 appeared. The Leica LAF 1200 range finder reported a range of 131 yards. Not a long shot but it would do for shot #1. Centering the reticle at the base of the chuck's tail as it walked up hill I fired. The .204s light recoil permitted a view of the bullet's impact.

The bullet had found its' mark and the chuck's body swelled as though pumped full of air for an instant before it simply collapsed. There was no kicking, flopping or chuck rendition of break dancing before giving up the ghost! Doc was watching the chuck while I made the shot and reported "Dead chuck" as the rifles report echoed.

Within a few minutes chuck #2 appeared and ducked back down into its' burrow. It was Doc's turn and he made ready with his Remington .223 VS. This chuck would peek out and drop out of sight in a flash. This was making it tough for Doc to get a good clean shot, I told Doc to make ready and I would make him appear. Doc shot me a look of disbelief, and asked if I was going to whistle him up. That old trick of whistling and the chuck stands up has long been played out as far as chucks are concerned in New York. In my 50+ years of hunting chucks I've seen it work but a few times, and not at all in the past 40 years. However, there is another tune to play that will make chucks follow the piper.

I have found blowing location, and feeding calls on a crow call will at times put chucks at ease,

and they will not be as reluctant to leave the safety of their borrow. While Doc made ready with his rifle I pulled an old Mallard Tone crow call from my daypack. and asked Doc if he was at ready. He gave a nod of yes and I gave forth a call in crow that said all is well and, I'm over here feeding. It wasn't but a few seconds and the chuck poked his head out to look around. Doc's 223 cracked and the bullet took the chuck through the head just below the ears. He folded up and slumped back in the hole! An old Indian trick I suppose Doc asked? No, just an old Italian crow hunters' trick! As the afternoon continued Doc killed two more chucks and I a total of three with the new Cooper. The .204 Ruger cartridge was proving to be accurate, and able to produce clean kills.

I made one more trip in early September of 2007 hunting chucks on a day the was unusually hot for that time of year. Knowing this would be the last trip of the 2007 chuck hunting season I opted to use the Cooper in .204 Ruger again to gain some more experience with the cartridge and rifle. Late in the afternoon I located a field that had been recently cut clear of hay. It appeared to me that the field was cut within a week as new growth was present. That stuff is woodchuck candy! They love it, and they just can't get enough of it this time of the year. High on that tiered hill, just below the ridge at the top, were located several chuck holes. They were easy to spot and on a prime location, a south slope. Although there were no chucks in sight I was sure this would be a hot spot for chuck activity late in the afternoon. The only problem was a lack of cover to hide and shoot from.

Using a few branches and some long grass from the edge of the field I erected a small hide on the lip of a tier a few hundred yards below the ridge where the chuck holes were located. There the reverse slope at the lip would allow me to lie prone with my daypack for a rifle rest in the cover of the branches and grass.

4:00 PM the sun was beating down on me as I lay prone in the hide watching the ridge. The hole on the far left looked very different when viewed again. Through the Leupold scope set at 20X I could make out the outline of a chucks head peeking out of its' hole. The hole was surrounded with grass camouflaging the chuck, but not good enough to hide from the Leupold. Using the Leica 1200 range finder I measured the distance to the chuck at 241yards. Elevation was set on the scope for the distance and rifle focused on the chuck. I waited for a few minutes before attempting to take the shot hoping that the chuck would expose more of itself. He ducked back into the hole, and poked his head back out again. That was when I let the shot go. In the scope I could see a plume of dust explode in a halo about the chucks head, and hear the sound of the bullets impact. One down!

That scenario was repeated four more times in the next hour as chucks would peek out of the holes located at the top of the hill.

It was nearly 5:45 PM when I made my way up the hill to verify the kills and inspect the results of the .204 Ruger's performance. All five chucks lie dead at the mouth of their burrow or slumped back into the hole. The little .204 bullet was doing a great job of hitting the mark and dispatching the varmint.

Looking back down the hill to where I had shot from another chuck was standing and looking at the branches I used to conceal my shooting position. It had no clue that I was behind it when the 204 cracked again. Chuck #6 dropped like a bag of rags to move no more.

There on that hill at that very moment I fell in love with the Cooper and the cartridge. It was no longer just a pretty rifle and a new interesting cartridge. What more could I ask for in a mid range varmint rifle and cartridge? I knew it was a keeper!

.204 RUGER LOAD DEVELOPMENTS

Over the remainder of 2007 a variety of .204 diameter bullet, suitable powders, and information was gathered in preparation for further research and load development of the .204 Ruger cartridge. The project goal was to develop loads that were at or near the factory published velocity, accurate and delivered effective terminal performance at the target up to 500 yards in distance.

Bullets and powder used in load development were selected from information found in loading manuals, and posting on web sites. AA2230, X-Terminator, XMR-2015, H322, BL-C#2, powders were chosen based on that information. There are several other powders listed in the loading manuals, but these five different powders represent a range of burning rates that appeared to be correct for the cartridge and bullet weights chosen.

Bullets selected for load development were chosen based on the twist rated of the Coopers rifling (1 turn in 12"). There is a wide selection of bullets and weights offered by several manufacture in the .204

diameter, but the twist rate of the rifle suggested bullets that weighed in the range of 32 to 40 grains as being correct at the velocities set as a project goal (3900 to 4100 fps).

Obtaining accuracy and velocity in this range of speed would give the 204 Ruger a substantial edge over other cartridges in the 223 Remington class. Not that there is anything wrong with the 223, but this would put the 204 Ruger in a niche that remains between the 223 Remington and the 22-250 Remington. The bullets selected were Hornady 40 grain V-Max, Berger 35 grain HP, Sierra 32 grain and 39 grain Blitz king.

Now rather than give you reader's twenty pages of test results from all the combination of powders and bullets tested I have rendered it down to three loads that came out perched on the top of the pile. They are the cream of the crop. They have achieved the goals set forth for accuracy, speed, and terminal performance at the target. However, before I just hand over this information there are a few details that you should be aware of.

#1 Acceptable accuracy was achieved throughout the testing with most all bullet and powder combinations. This in part is no doubt due to the fact that Cooper builds a fine rifle.

#2 Achieving the set velocity goals with acceptable pressures limits was only accomplished with one powder tested. The other powders tested produced velocities in the range of 3500 to 3600 fps when pressures sign started to become an issue.

#3 Top accuracy was only achieved when the bullet was ***HARD SEATED IN THE RIFLING.**

*For those of you that are not familiar with **HARD SEATING a BULLET** I will explain the concept. **Hard seating a bullet** is simply seating a bullet in the case so that the bullet makes contact with the rifling lead when the cartridge is fully chambered in the ready to fire position. When properly seated for a HARD SEAT the rifling lead ramps will leave small square contact marks around the circumference of the bullets ogive where the bullet and the rifling lead ramps mate.

Seating a bullet in this manner to improve accuracy is not a new concept. In fact it is well established more than 100 years ago when the target shooters of the late 1800's breach seated lead bullets they shot more accurately. This method helps to eliminate most bullets to bore misalignment problems as Dr. Mann found and proved in his experiments more than 100 hundred years ago.

Here are a few tips that you should know before you try **HARD SEATING** bullets.

#1 DO NOT USE A LOADED CARTRIDGE TO MAKE SEATING DEPTH ADJUSTMENTS

When establishing the proper seating depth to achieve a HARD SEAT never use a live cartridge. There are far too many things that can go wrong, an accidental discharge, the bullet becoming lodged in the bore, and so on. Use an un-primed case without powder to make a dummy seating depth gauge.

#2 HARD SEATING a bullet will increase chamber pressure, so reduce the powder charge and work up in half grain increments.

#3 A bullet, when properly HARD SEATED, will show bullet to rifling lead ramp contact points on the bullet's ogive completely around the bullet's circumference. The contact marks will appear as small squares, and not rectangles. If the marks left are rectangles that are along the axis of the bullet it is seated too far into the rifling. This will pose two problems. The first is higher chamber pressure. The second is a bullet that may become lodged in the rifling and remain there when a loaded cartridge is extracted. With the bullet stuck in the rifling the powder will be dumped in the chamber and bolt recess of the action. Now you've a mess on your hands that will take a cleaning rod to dislodge the bullet, and you will need to clean the un-burnt powder from the chamber and action. This is something that you don't want to have happen in the field or in the middle of a match.

The three loads developed that achieved all three of the goals set in load development are as follows. They are at or near maximum loads and should not be attempted as a starting load. **Rather reduce the suggested load by 10% and work up in half grain increments. I can't stress the importance of this enough as I**

found in testing that as little as .2 grains can make a big difference in this cartridge when nearing a maximum load.

*Velocity corrected using Miller formulae for distance from muzzle to center of chronograph screens. SD from second string of 5 rounds fired through a barrel with 5 previously fired shots with the same load. The Cooper 21 barrel is 26" long, and has a 1 in 12' twist rate. Oehler 33 chronograph used to record velocity.

Load #1

Sierra 39 grain Blitz King Bullet
Winchester case
Federal 205M primer
BL-C#2 powder at 30.2 grains
*Velocity average 3949 fps, SD of 5.2 fps
Accuracy at 200 yards for 5 shots .460" or .230 MOA

Load #2

Sierra 32 grain Blitz King Bullet
Winchester case
Federal 205M primer
BL-C#2 powder at 30.4 grains
*Velocity average 4079 fps, SD of 6.1 fps
Accuracy at 200 yards for 5 shots .691" or .345 MOA

Load #3

Berger 35 grain Match/Varmint HP
Winchester case
Federal 205M primer
*Velocity average 3999 fps, SD test data lost
Accuracy at 200 yards for 5 shots .418" or .209 MOA

These three loads achieved the goals set forth at the start of the load development project. The rifle and ammo were now ready for field testing while varmint hunting in 2008.

2008 .204 RUGER FIELD TEST RESULTS

From May through mid September of 2008 I carried the Cooper 21 in .204 Ruger exclusively while varmint hunting. Woodchucks were the primary target species, but crows and wild pigeons were not passed up when the opportunity presented itself.

Load #1 with the 39 grain Sierra BK bullet turned out to be the best all around load. It produced quick, clean kills on chucks at ranges out to 358 yards. In most all cases the bullet never exited the woodchuck and dropped them right where they were shot.

Wild pigeons were encountered at a range of 241 yards in a cut corn field. When several of them lined up one behind the other the Cooper sent them a 39 grain bomb. On bullet impact there were pigeon parts and feather in the air, but I could not tell how many had been hit. The remainder of the flock lifted off and I kept an eye on them as they flew away. One member of the flock was observed experiencing apparent engine trouble and made a limp spinning fall to the ground. When I reached the place where the pigeons were shot I found an area of about 16 feet in diameter littered with feathers and parts. It took several

minutes of gathering parts, and adding them to come up with a body count of 5 birds. At that point the .204 Ruger was declared a 6 pigeon rifle.

Crows shot with this load spectacularly exploded like a quarter stick went off in a bag full of black crepe paper. To say that the Sierra 39 grain Blitz King bullet exhibits explosive expansion is pure fact.

I'd like to make mention of a test that Harold Miller and I conducted using the Sierra 39 grain Blitz King bullet in his Borden built 20 Vartarg. Primarily I make mention of this as testimony to the accuracy exhibited by this bullet.

While Harold and I were working together on our projects he asked me to fire a group of 5 shots with his rifle and ammo at a range of 500 yards. I suggested that he work with me while I fired the group serving as an observer of wind and condition. That way I could concentrate on sight picture and trigger while he would report the condition as right with a command to fire. Harold watched the conditions very closely and the target with a 40X Leupold spotting scope. He did not report shot location while I fired but rather remained silent till conditions were right and then would give the command to fire.

After firing shot #5 in the string I looked over at Harold who had a broad smile. "Let's take a walk out to 500 for a closer look at the group" he said. When we arrive at the target I was amazed at the group. It had printed a few inches low of the point of aim but had placed all five Sierra Blitz King bullets in a group that was slightly over 2 1/2" horizontally and under 2" vertically. This is great performance for any bullet, but a tiny .204 diameter one that weighs only 39 grains accomplishing this is truly outstanding.

Load #2 with the 32 Grain Sierra Blitz King bullet proved to be even more explosive than its 39 grain sibling. I was presented with a shot at a large female woodchuck at a range of 87 yards. From a setting position I placed the bullet just behind the chuck's shoulder. My view through the Leupold set on 20X was not obscured by the recoil and I could see the bullet explode on impact. The entrance wound was a hole of nearly 2 1/2" in diameter and there was no exit. The chuck just dropped where it was shot. For large chucks the 32 grain BK maybe a little too explosive, but for smaller varmints like prairie dogs it should be just great!

Load #3 Berger 35 grain Match/Varmint HP bullet proved to be less explosive than the Sierra Blitz King bullets. It did leave exit holes in woodchucks. I believe it would make a great bullet for varmints larger than woodchucks. It also proved to be a very accurate bullet, and one I'd surely to use in a match.

A total of 40 woodchucks, 3 crows and 9 wild pigeons were shot with the Cooper using the three loads afore mentioned. Although all three loads got the job done I have settled on load #1 for my basic all around varmint hunting load in the .204 Ruger. It has proven to be very accurate, has the highest BC, and great terminal performance on live targets.

The testing and load development research mentioned within was extensive, however it is not to be thought of as conclusive in load development for this fine cartridge. Rather it is intended to help those of you who may wish to take up the .204 Ruger and experience what it has to offer.

May all your shots find their intended

mark, John Del Savio



Picture here the crew is taking a lunch break, left to right Harold Miller, Fran Plew, and Dr. Richard (Doc) Maretzo). Their enthusiasm for the sport is apparent in that they are on break and still scanning the fields for chucks. Note Doc with the spy glasses rather than a sandwich. Note the large chuck in front of Fran Plew's feet. It's not sun bathing.



Using an ATV while chuck hunting the author can hunt/scout more area then can be covered on foot. Late summer woodchucks pack on the weight in preparation for a long winter of hibernation as can be seen in the photo. These 9 were spared the ordeal of that long winter underground via .204 diameter bullets.



Here the author holds two woodchucks he took with the Cooper 21 in .204 Ruger. The chuck on the left was taken with a Berger 35 grain Match/Varmint bullet used in Load #3. Note the exit hole. That chuck was peeking from the presumed safety of its' clover camouflaged borrow. Even at a range of nearly 300 yards the Leupold 6.5-20X sees all. The little 35 grain Berger bullet at 3999 fps found its' mark and proved to be accurate and deadly.



For those of you that are not familiar with Woodchucks here's a little information about this varmint that may lead to an understanding of why they are hunted.

MARMOTTA MONAX Mamalis, herbivore, largest member of the squirrel family. (a.k.a. woodchuck, chuck, ground hog, pasture poodle, and whistle pig). Native to central, and eastern North America east of the Mississippi. Average adult 18" tall and up to 20 pounds. Usually has a litter of pups once a year ranging from 4 to 6 offspring sometime in early spring. Burrowing ground dweller that hibernates from late fall till early spring.

This cute looking fuzzy creature has basically four activities through out its' life, borrowing, hibernating, reproducing and eating everything in sight that's green. It has been estimated that a woodchuck eats approximately $\frac{3}{4}$ of a ton of crops per year. Do the math! One adult male, one adult female, six pups, that will reach adulthood in one season, = six tons of crops depleted from the farmer's harvest per year. It's easy to see the economic impact to the farmer.

Add to the woodchuck insidious appetite a constant behavior of burrowing. Prior to north America becoming agricultural woodchucks were forest inhabitants. Once land was cleared and crops grown the abundance of food was more than they could resist. Moving their burrows to farm land was woodchuck heaven! The tilled soil was easy to burrow in and the food was right there on their doorstep.

Pastures having chuck burrows are kind of a livestock / farm machinery mine field. Valuable cattle, and horses are at a high risk of breaking a leg in chuck holes. The burrows also accelerate erosion, and cave in when farm equipment is driven over them. Many a farmer has had his farm machinery damaged, suffered injury, or even loss of life as the results of woodchuck burrows. Thus started the great North American Farmer versus Woodchuck War that still rages today.